Emergence of Intrinsic Length Scales in Active Cytoskeletal Assemblies

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Background and Significance

Unlike materials at equilibrium, active matter systems are composed of nonlinearly interacting components that continuously consume energy to generate motion. This activity produces complex dynamics and organization of flows and structures which can be seen in examples across a wide range of scales from cytoskeleton filaments and molecular motors to populations of motile organisms [1]. Understanding how collective organization arises in these systems is a major challenge in nonequilibrium physics and may reveal how microscopic interactions drive organized motion in biological systems.

Microtubule-kinesin networks serve as controllable experimental systems for examining these emergent behaviors [2]. Microtubules (MT) are cytoskeletal filaments, while kinesin proteins are motors which use chemical energy from ATP to move along microtubules and generate force between them. A fundamental question in these systems is what determines the characteristic spatial scale of collective motion, measured here using the velocity correlation length, which describes the distance over which motion in the material remains correlated. Because active systems self-organize, controlling this characteristic length scale through external parameters is generally more difficult than in passive systems. Literature has shown that as confinement is reduced by increasing chamber height, the correlation length increases before saturating at an intrinsic length scale [3]. This suggests that when confinement becomes sufficiently weak, the characteristic scale is determined by interactions within the active material rather than external system boundaries [4].

In this project, we introduce a second, passive entangled filament network into the active material as a potential way to modify the characteristic length scale of motion within the system. Our preliminary experiments suggest that interactions with this passive network can alter the organization and spatial scale of the active flows. My goal is to determine whether the resulting length scale is truly intrinsic to the material, meaning that it becomes independent of the geometry of the confining chamber. Establishing this would help reveal how interactions within active materials can determine their organization independently of external boundaries.

Specific Aims, Goals, or Questions

The primary goal of this project is to determine when and how an intrinsic length scale independent of geometric confinement emerges. We hypothesize that increasing the concentration of passive filaments causes the system to shift from confinement-dominated behavior, where the velocity correlation length varies with chamber height, to behavior in which interactions within the active network establish a characteristic length scale across all chamber heights.

To test this hypothesis, I will systematically vary the properties of the passive filament network and compare the resulting collective flows across different chamber heights. This will allow me to determine the conditions under which the system transitions from confinement-dependent behavior to behavior in which an intrinsic length scale emerges from internal interactions. A further goal is to determine whether this transition depends only on the amount of passive filaments present or also on its physical structure.

Another question to investigate is the time required for an intrinsic length scale to emerge after activity starts. Preliminary results at higher passive filament concentrations suggest that thicker chambers may take longer to approach the characteristic length scale. The mechanisms controlling this timescale are currently unknown, and we aim to investigate what causes this delayed emergence.

Approach

To investigate how geometric confinement affects these systems, I will prepare experimental chambers with several different heights. The chambers will be fabricated from a transparent silicone material and plasma bonded to glass coverslips to create sealed environments for imaging. Plasma treatment modifies the surfaces of the silicone and

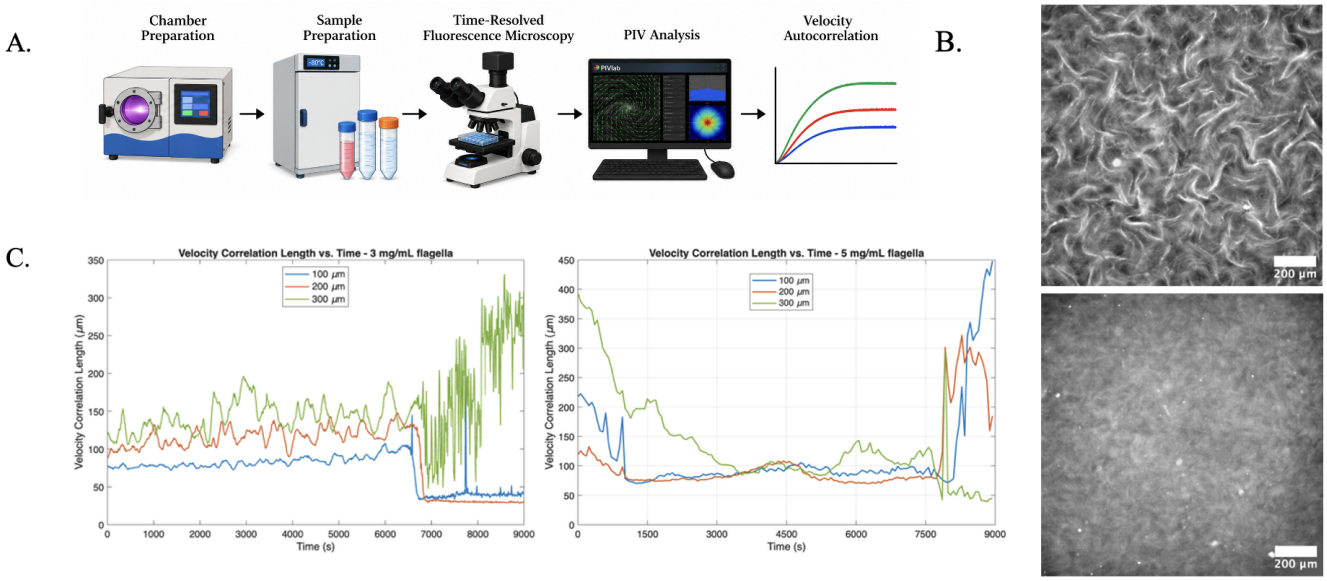


Figure 1: A. Abstract schematic of work flow and key instrumentation and computational tools to be used in proposed work (generated using ChatGPT-5.6 Sol). B. Fluorescence microscopy images of an active microtubule network without the passive filament network in a 100 μm chamber (top) and with the passive filament network in a 100 μm chamber (bottom) (author's original work). C. Velocity correlation length as a function of time across three chamber heights for two different passive filament network concentrations (author's original work).

glass, allowing them to form a strong covalent bond that prevents leakage. The chambers will then be treated with a passivating agent to minimize unwanted interactions between the active material and the chamber walls.

I will prepare samples containing an active network of microtubules, motor proteins, and crosslinking proteins together with a passive network of polymerized bacterial flagella, which are stiff protein filaments that enable bacterial movement and can adopt different structures, including straight and helical configurations. In the active network, the bundling protein PRC1 crosslinks overlapping microtubules, allowing kinesin-streptavidin assemblies (KSA), which are clusters of motor proteins, to drive their motion [5]. I will systematically vary the concentration and polymerization conditions of the passive filament network to determine how its concentration and physical structure influence collective motion, as polymerized flagella can adopt different structures depending on environmental conditions [6].

Prepared samples will be introduced into chambers of different heights and imaged using time-resolved fluorescence microscopy throughout their period of activity. The images will be preprocessed to improve brightness and contrast, and flow velocities will be measured using particle image velocimetry (PIV), a technique that tracks patterns between successive images to determine the direction and speed of local motion. From these velocity fields, I will calculate the root-mean-square velocity and velocity correlation length over time. Comparing these measurements across chamber heights and passive network conditions will determine when the characteristic length scale becomes independent of confinement and how that intrinsic scale depends on the properties of the passive network.

Outcomes

Understanding how internal material properties and external confinement control collective organization in active matter may ultimately contribute to designing systems with adjustable dynamic behavior with potential applications in targeted drug delivery, smart materials, and soft robotics [7]. My work will directly contribute to an ongoing manuscript investigating how passive flagella networks influence the activity and characteristic length scales of PRC1-KSA active matter. My experiments will provide the manuscript with a direct comparison between confinement-dependent and intrinsic length-scale behavior, helping establish how flagella concentration, structure, and chamber geometry collectively determine active network organization. Through this project, I will also develop greater independence in experimental design, microscopy, quantitative image analysis, and scientific communication while contributing original results to an ongoing research effort.

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